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Corporate communication and likelihood of data breaches

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ABSTRACT

We use data breaches of U.S. firms from 2005 to 2018 to examine the association between corporate communication and the likelihood of data breaches. We find that positive tone of annual reports (10-K filings) has a significant and positive association with the odds of a data breach. We further explore the related mechanisms and find strong evidence that the association between positive tone of annual reports and odds of a data breach is largely attributable to a firm's internal resources and market prospects (proxied by 'cash to total assets' and 'excess stock returns'). We also find that in the firms with more readable 10-K filings, positive tone is not associated with a higher likelihood of a data breach. This finding highlights the importance of communication tone and transparency in the context of potential data breaches.

1. Introduction

Data breaches and cyberattacks[1] have become increasingly ubiquitous over the years. News headlines and highlights are not only showing the large number of records compromised (e.g., "Warning As 26 Billion Records Leak", reads a Forbes headline on January 23,2024[2]), but they are also highlighting attacks that are becoming increasingly sophisticated and cross-border in nature by concurrently affecting organizations in multiple countries (e.g., a 2023 MOVEit cyberattack affected more than 65 million individuals and more than 2300 organizations in 30 countries, leading to more than $10 billion in global cost (Madnick, 2023)).

The near daily occurrences of data breaches have significant financial consequences for organizations - in the form of both direct costs (e.g., lost sales resulting from cyber-attacks) as well as indirect costs (e.g., reputational damage) (McAfee, 2014). The New York Times (2017), for example, notes that Verizon had to shave off $350 million (i.e., an over 7% discount) from Yahoo's purchase price because of unknown costs related to Yahoo's data breaches. CNN (2022) also reports that Meta had to pay $725 million to settle a lawsuit over a (Cambridge Analytica) data leak. Similar stories abound in the popular press concerning the costs sustained by companies due to data breaches. Equifax's data breach led to a 14% drop in its share price in one day (Wall Street Journal, 2017) and a $70

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[1] Given the proximity of their meanings, in this study we have used the terms 'data breaches', 'cyberattacks' and 'hacking' interchangeably.

[2] See https://www.forbes.com/sites/daveywinder/2024/01/23/massive-26-billion-record-leak-dropbox-linkedin-twitterx-all-named/?sh=487ac48fab58.

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billion class action lawsuit (Bloomberg, 2017).[3] Also, following a 2021 data breach, T-Mobile had to pay $350 million and additionally spend $150 million to upgrade data security in order to settle litigation over that cyberattack (Reuters, 2022). Indeed, Cybersecurity Ventures (2024) estimates that cybercrime will cost the world economy $9.5 trillion in 2024. A number of academic studies have found the negative effects of data breaches to include loss of market value and deterioration in operating performance. For instance, Garg, Curtis, and Halper (2003), Cavusoglu, Mishra, and Raghunathan (2004) and Kamiya, Kang, Kim, Milidonis, and Stulz (2021) have reported significant negative stock market reaction following the announcement of data breaches. As well, Kamiya et al. (2021) find that data breaches lead to decline in operating performance while Rosati et al. (2017) find the bid-ask spread to widen on the day that data breaches are announced. Further, Lattanzio and Ma (2023) find that data breaches lead to declining corporate innovation.

Despite the pervasiveness of data breaches, their burgeoning costs and the continuing threats they pose, the literature on what makes a firm more vulnerable to data breaches is still developing. Among the studies, Kamiya et al. (2021) use accounting and governance data (considered as 'hard-facts') and find that firms that are more profitable and have more future growth opportunities are more susceptible to data breaches. The empirical evidence that resourceful and profitable companies are more attractive to cyber criminals is consistent with the predictions of fraud theory (Cressey, 1950; Wolfe & Hermanson, 2004). Since profitable firms with better financial prospects could mean more financial benefits if cyberattacks are successful, cyber criminals can rationalize the launch of cyberattacks on resourceful and profitable companies on the grounds that the potential financial benefits of a successful attack are worth the risks, thereby increasing the probability that such companies will experience data breaches. However, earlier studies have pointed out that, in addition to the 'hard-facts', one should also focus on the 'soft information' of a firm to have a deeper insight on its future prospects (Tetlock, Saar-Tsechansky, & Macskassy, 2008). Such utility of 'soft information' may encourage data breachers[4] to pay a closer attention to the information embedded in corporate disclosure. This possibility is supported by the observation of Bonsall, Leone, Miller, and Rennekamp (2017) that "A growing body of research in accounting and finance examines whether, and to what extent, qualitative attributes of corporate communication (e.g., tone, readability) affect the decision-making of investors and information intermediaries" (p. 329). Given that data breachers' main objective is to steal valuable and private information of a firm - in addition to the 'hard-facts' - the nature of firm disclosure and its informativeness is also likely to play a role in data breachers' unwarranted actions. Yet, to the best of our knowledge, no prior studies have pursued any systematic inquiry into this intriguing research question. Accordingly, our primary research question is: *does the nature of corporate communication (*e.g.*, positive/negative, litigious, constraining tone) have any significant association with the likelihood of a data breach?* We also focus on a complementary research question: *How does the readability of 10-K reports (i.e., clarity of 10-K information disclosure) affect the relation between 10-K tone and data breach likelihood?* Since these questions have not received much attention in the literature, our study aims to fill the gap.

More specifically, we focus on the positive tone of 10-K reports (Loughran & McDonald, 2011)[5] – the most important and comprehensive corporate communication medium – and the readability of such reports. Firms with more positively toned reports are likely to draw greater attention of hackers and data thieves – because positive reports strengthen the 'resourcefulness' tag of a firm. Positive tone in 10-K report may signal good prospects for a firm, which would make data breachers or hackers more interested in the firm in order to make pecuniary gains. Relatedly, firms have a choice of presenting their mandatory 10-K disclosures in simple or complex terms. While SEC (1998) touts simplicity and encourages companies to make disclosures in plain and simple to read English, a number of companies still choose complexity in their disclosures. After forty years of studying documents filed by public companies, legendary investor Warren Buffett admits that, too often, "I've been unable to decipher just what is being said or, worse yet, had to conclude that nothing was being said" (SEC, 1998, p. 1). We argue that a firm's informational complexity (e.g., lower readability of narrative disclosures) may increase the probability of being hacked. Since lower readability makes it difficult for market participants to retrieve information from corporate communication (e.g., 10-K reports) (Lehavy, Li, & Merkley, 2011), it might make the hackers or data breachers more '*curious*' about the firm and motivate them to hack or breach a firm's informational domain. In other words, we contend that tone and readability of 10-K disclosures will potentially have an influence on the likelihood of being hacked.[6]

Given the need for, but absence of evidence on these important issues, our study is intended to fill these gaps in the literature. Using the sample of data breaches of U.S. firms reported in the Privacy Rights Clearninghouse (PRC)[7] from 2005 to 2018, we confirm the significance of cyberattacks by showing that data-breaches are associated with significant and negative cumulative abnormal returns (CAR) around the event dates. Subsequently, consistent with our expectations, we find that the positive words ratio in 10-K filings (i.e.,

[3] CBC notes that Equifax shares fell 13.7% on the day the breach was made public, tumbling $19.49 US to close at $123.23 (http://www.cbc.ca/news/business/equifax-breach-1.4280682).

[4] 'Data breachers' terminology is not formally used in English literature. By 'Data breachers' we refer to the persons who take part in data breaching and data theft activities.

[5] We focus on the 10-K reports, which are considered to be the most important and comprehensive corporate narrative disclosures and are analyzed very commonly in the financial and accounting literature (Loughran & McDonald, 2011).

[6] Lower readability may also signal weaker protection against cyberattacks. Earlier results show that readability is related to managerial ability (Hasan, 2020) and corporate governance (Chakrabarty, Seetharaman, Swanson, & Wang, 2018). Highly complex and less readable disclosure may signal less sophisticated management or poor governance practices in a firm. Firms with such attributes may be unable to protect themselves from the onslaught of sophisticated and determined cyber-criminals. On the other hand, simple and easy to read disclosure may signify sophisticated management and/or better governance practices capable of outwitting determined hackers.

[7] Privacy Rights Clearninghouse (PRC) is a not-for-profit organization that, among other things, maintains a chronology of data breaches as a source of information to assist in research involving reported data breaches from 2005 to present. PRC defines a "data breach as a security violation in which sensitive, protected or confidential data is copied, transmitted, viewed, stolen or used by an unauthorized individual". For more information, visit https://www.privacyrights.org/data-breaches.

management presenting a more positive or resourcefulness image of their company) is significantly and positively related to the odds of experiencing a data breach after controlling for "hard facts" of financial resources and performance reflected in financial ratios. The positive image in the disclosures possibly signals good financial resources and future prospects, thus, potentially making it enticing for fraudsters (i.e., hackers and data breachers) to launch cyberattacks on the firm. Our empirical analysis supports this mechanism by showing that the association between positive tone of annual reports and odds of a data breach is largely attributable to a firm's internal resources and market prospects (proxied by 'cash to total assets' and 'excess stock returns'). We also find that companies that make their disclosures using complex and less readable language are more likely to be breached. Quite interestingly, we further find that in the firms with more readable (i.e., less complex) 10-K filings, positive tone is *not* associated with a higher likelihood of data breach. It appears that a higher level of transparency in narrative disclosures can be effective in quenching the curiosity of hackers and data breachers and can make them less interested in launching a cyberattack on the focal firm. This finding highlights the importance of communication transparency that mitigates the unwarranted inquisitiveness of data breachers and hackers. These results offer some directions on how to improve readability of firm disclosures, which in turn can reduce data breach probability.

While our study presents some insightful results, these are susceptible to the common concerns of endogeneity prevalent in corporate finance research – such as reverse causality and omitted variable bias. In other words, there is a possibility that firms adjust their narrative disclosures in anticipation of future cyberattacks (i.e., reverse causality). It is also possible that some characteristics such as firm performance may affect 10-K positive disclosure as well as data breach probability – posing a classic omitted variable problem. Admitting the challenges to address these concerns, we have controlled for most of the important determinants of data breaches as identified in the literature (e.g., Kamiya et al., 2021). However, since data breaches are relatively rare events, the lack of within-firm variation for this variable does not provide sufficient power to conduct firm fixed effect analysis to control for other unobservable time invariant firm characteristics (Zhou, 2001). Similar to Kamiya et al. (2021), we use pooled sample regression with robust standard errors clustered by firm and control for year and industry fixed effects in our models. While our empirical set-up alleviates the endogeneity issues to some extent,[8] yet we employ several additional approaches to alleviate the relevant concerns. Firstly, we use instrumental variables (IV) regression to address the endogeneity concerns related to both omitted variables and reverse causality issues. Following Ertugrul, Lei, Qiu, and Wan (2017), we adopt the mean value of positive words ratio from firms located within the same area in a given year as our instrument. In addition, we employ a matching firm analysis to address concerns about potential biases from other observed confounding factors. After creating a balanced sample based on all covariates we use in the baseline model, the treatment effect remains positive and significant. To further alleviate the concern of reverse causality, we repeat our analysis after excluding internal data breaches and we obtain qualitatively similar results. Our results remaining robust in these empirical settings helps to strengthen the validity of our findings by alleviating endogeneity concerns.

Subsequently, we explore the potential mechanisms that explains the positive relation between 10-K positive disclosure and data breach probability. We find strong evidence that the association between positive tone of annual reports and odds of a data breach is largely attributable to a firm's internal resources and market prospects (proxied by 'cash to total assets' and 'excess stock returns') - implying that positive tone reinforces the fraudsters resolve to attack the firms with more resources and future prospects.

Our paper makes some important contributions to the growing body of data breach literature in the following ways. The current literature on data breaches primarily focuses on the short- and long-term performance implications of data breaches (e.g., Kamiya et al., 2021); the costs, benefits and optimal investment in mitigating cyberattacks (e.g., Gordon & Loeb, 2002; Krutilla, Alexeev, Jardine, & Good, 2021; Paul & Zhang, 2021), cybersecurity disclosure (Chen, Henry, & Jiang, 2023; Florackis, Louca, Michaely, & Weber, 2023; Jiang, Legoria, Reichelt, & Walton, 2022), and cybersecurity and audit/examination services (Calderon & Gao, 2020; Cheong, Yoon, Cho, & No, 2021; Perols & Murthy, 2021). Our study differs from existing literature in important ways as we primarily focus on the impact of corporate communication on the likelihood of data breaches. For example, we use the popular tone of mandatory 10-K disclosures (Bodnaruk, Loughran, & McDonald, 2015; Loughran and McDonald, 2011, 2016) and widely used readability measures (namely the Fog Index and the Flesch-Kincaid Index) of 10-K reports to assess the chances of experiencing data breaches or cyberattacks.

Our study also makes contribution to the media and corporate communication literature in the following ways. Sentiment (i.e., tone) literature on media and corporate communication primarily focuses on negative tone or sentiment. Earlier studies (Loughran & McDonald, 2011; Tetlock, 2007; Tetlock et al., 2008) show that negative media coverage matters the most in terms of affecting investors' opinion, stock returns and trading activities, while articles with a positive tone are largely ignored and discounted. In this study, we show that positive tone also matters in specific contexts. Similarly, our study adds to the readability (or corporate communication complexity) literature as well. "Research examining the consequences of 10-K reports' readability shows that readability is positively associated with the earnings persistence (Li, 2008), analyst coverage, accuracy of forecasts (Lehavy et al., 2011), credit rating (Bonsall & Miller, 2017), stock liquidity and trading volume (De Franco, Hope, Vyas, & Zhou, 2015; Lang &

[8] First, cyber-attacks (or data-breaches) are random and mostly external events. Therefore, it is unlikely that firms will prepare their 10-K reports in anticipation of any future attack(s); rather firms will focus on their information system securities and relevant governance mechanisms. Second, firms prepare their 10-K reports for different stakeholders – such as investors, regulators, rating agencies, and financial analysts. Keeping in mind the expectations of diverse stakeholders, firms are likely to use more positive tone and less complex language (i.e., high readability) to make the firm more attractive and informationally efficient. However, as our empirical results show, these strategies (especially the positive tone disclosure) are not always effective in avoiding future data breaches. Third, given that textual variables are measured in the year before data breach, it is unlikely that reverse causality can drive the results. Particularly, prior research suggests that firms do not frequently make changes in annual report texts (e.g., Cohen et al., 2020).

Stice-Lawrence, 2015; Miller, 2010), and investment efficiency (Biddle, Hilary, & Verdi, 2009), while it is negatively associated with the cost of debt and the stock price crash risk (Ertugrul et al., 2017; Hasan, 2020, p. 150). Habib and Hasan (2020) also demonstrate that annual report readability is significantly associated with business strategy, while Zhang, Zhou, Pan, and Jia (2019) find that cryptocurrency traders respond to more readable authority-related news. We add to this literature by showing that readability of corporate disclosures is also an important determinant of data breaches/cyberattacks.

The rest of this paper is structured as follows. Section two is devoted to a review of the relevant literature and hypotheses development while section three discusses the data and methodology. The results are presented in section four; section five presents additional tests results. Section six concludes the paper.

2. Literature review and hypotheses

2.1. Implications of data breaches

Data breaches have become increasingly rampant over time (Buttrick et al., 2016). In the U.S. alone, the Privacy Rights Clearninghouse documented 22,060 data breaches among for-profit businesses between January 2002 and September 2023, which averages to about 1050 breaches per year or about just under 3 data breaches every day among U.S. for-profit companies. The breaches are affecting companies of all sizes and in all industries including almost all the tech giants (e.g., Microsoft, Google, Facebook, Apple) and even companies whose "bread and butter" are protecting the privacy of our information (e.g., Equifax). As well, the data breaches are becoming increasingly massive based on the number of records compromised. Forbes (2024) notes that 26 billion records were leaked in a "mother of all broaches" (MOAB). Data compiled from various websites (Fig. 1) gives a synopsis of the number of records breached in some of the biggest publicly disclosed breaches up until March 30, 2024. From Fig. 1, the 2024 "mother of all breaches" which leaked 26 billion records tops the list while the 2021 CVS breach which led to the exposure of a billion records rounds off the top 10 big breaches.

Data breaches can be detrimental to the long-term financial performance of a company, especially since customers whose sensitive information has been compromised can lose confidence in a company's ability to safeguard their sensitive information, which can negatively affect a company's brand and sales. Thus, the costs of data breaches can be significant and can include direct costs such as payments made in ransomware demands, cost incurred to clean up the data breaches, lost customer sales as well as indirect costs such as lost employee time or productivity resulting from information systems interruption. Ponemon Institute (2023) finds that the average elapse time to identify and contain a data breach is about 277 days and the average cost of a breach is $4.45 million globally and $9.48 million in the U.S.

Several other studies have examined the costs of data breaches and find the costs to be significant. Studies examining the negative effects of data breaches on market value and stock price performance include Garg et al. (2003) who examine 22 IT security breaches from 1996 to 2002 and find a 2.7% fall in stock price over one day and 4.5% decline over a three-day window. Also, although Campbell, Gordon, Loeb, and Zhou (2003) find limited evidence of overall negative stock price reactions to the data breaches they studied, the authors find that data breaches involving confidential access to data garnered significant negative stock market reaction while those that did not involve accessing confidential data did not lead to any significant stock price reaction. Cavusoglu et al. (2004) find that breached firms, within 2 days of an announcement of Internet security breach, lost an average of 2.1% of their market value,

Name	Number of Records Exposed	Year
Mother of all Breaches (multiple breaches compiled)	26.0 Billion	2024
Compilation of Many Passwords (multiple companies)	8.4 Billion	2023
AP Network (Airplane Accelerates app)	5.7 Billion	2022
DarkBeam	3.8 Billion	2023
Compilation of Many Breaches (multiple companies)	3.2 Billion	2022
Yahoo	3.0 Billion	2013
Orvibo	2.0 Billion	2019
River City Media	1.3 Billion	2017
Alibaba	1.1 Billion	2019
CVS	1.0 Billion	2021

Fig. 1. Largest publicly disclosed data breaches so far (up to March 30, 2024)
This figure, compiled by the authors based on various websites, shows the largest number of records compromised so far in data breaches (i.e., up to March 30, 2024) that have been reported on various websites (retrieved last on April 19, 2024). The websites are: 1) https://www.forbes.com/sites/daveywinder/2024/01/23/massive-26-billion-record-leak-dropbox-linkedin-twitterx-all-named/?sh=487ac48fab58; 2) https://cybernews.com/security/rockyou2021-alltime-largest-password-compilation-leaked/; 3) https://cybernews.com/security/5-7bn-data-entries-found-exposed-on-chinese-vpn/; 4) https://www.idstrong.com/sentinel/darkbeams-alarming-data-breach/; 5) https://cybernews.com/news/largest-compilation-of-emails-and-passwords-leaked-free/; 6) https://www.nytimes.com/2017/10/03/technology/yahoo-hack-3-billion-users.html; 7) https://www.forbes.com/sites/daveywinder/2019/07/02/confirmed-2-billion-records-exposed-in-massive-smart-home-device-breach/?sh=4dfaa65e411c; 8) https://www.theguardian.com/technology/2017/mar/06/email-addresses-spam-leak-river-city-media; 9) https://www.itpro.com/security/data-breaches/359897/alibaba-data-breach-exposes-11-billion-pieces-of-data; and, 10) https://www.fiercehealthcare.com/tech/cvs-health-database-leak-leaves-1-billion-user-records-exposed-online.

which translates to $1.65 billion loss in market capitalization per breach. Lending, Minnick, and Schorno (2018) find that firms that have experienced data breaches have one-year buy-and-hold abnormal returns of about −3.5%. Kamiya et al. (2021) find significantly negative cumulative abnormal return of −0.84% over a three-day window around a cyberattack announcement, which translates to an average loss in company value of $495 million per attack. Florackis et al. (2023) and Jiang, Khanna, Yang, and Zhou (2024) find their measures of cybersecurity risk to be priced in the cross-section of stock returns, with portfolios of firms having high exposure to their cybersecurity measures earning higher average stock returns.

Data breaches also have negative impact on the operating performance and other aspects of affected companies. Lending et al. (2018) find that a data breach leads to significant declines in deposits and other commercial activities in banks. Kamiya et al. (2021) report significant declines in sales growth and credit ratings and increases in bankruptcy probability of affected firms. However, Ko and Dorantes (2006) find that data breaches have minimal long term economic impact on affected companies. Rosati et al. (2017) examine 74 data breaches that occur between 2005 and 2014 and find an increase in bid-ask spread and trading volume on the day of announcement of data breaches. Data breaches can also lead to reputational loss (Kamiya et al., 2021) and may increase the likelihood of chief executive officer and chief technology officer turnover (Lending et al., 2018). The costs of data breaches are also felt in declining innovation and competitiveness. As Lattanzio and Ma (2023) find, firms that are faced with increased cybersecurity risk are relying less on trade secrets and filing simple patents to get protection under patent laws, which is leading to declining returns on R&D investments. In general, empirical evidence demonstrates that data breaches impose significant costs on almost all facets of affected companies.

Going forward, data breaches are expected to become even more rampant, costly and sophisticated as we enter the artificial intelligence (AI) era (Fortune, 2024; MIT Technology Review, 2023), as bad actors (PC Gamer, 2024), aided by AI, especially deceptive AI systems that have developed the ability to hide their unsafe behavior and are capable of eluding standard techniques designed to remove deceptive behavior (Hubinger, Denison, Mu, & Lambert, 2024), will potentially unleash data breach nightmare scenarios for companies. For example, cyber swindlers tricked a finance worker in the Hong Kong branch of a multinational firm into paying $25.6 million to them by using deepfake technology to pose as the firm's CFO in a video conference call (Business Insider, 2024).

2.2. Relevant theory: data breach and corporate communication

With 94.6% of data breaches being financially motivated (Verizon, 2023), data breaches are largely financial crimes driven by the desire for financial gain. As financial crimes, fraud theory (i.e., the fraud triangle theory and fraud diamond theory) as well as recent enhancements to fraud theory, provide the theoretical underpinning and motivation for our study. Fraud theory posits that the confluence of financial pressure or incentive or motive to commit a crime, the opportunity to commit the crime, the rationalization of the fraudulent behavior and capability to commit the crime – drive the commission of financial fraud (Cressey, 1950; Wolfe & Hermanson, 2004). If cyber criminals see that the financial benefit they will derive from successful cyberattacks on a company are high, they can rationalize launching cyberattacks on the basis that it is worth taking the risk because of the high potential financial gain. Firm financial prosperity can be inferred from quantifiable financial metrics (such as higher cash reserve, higher market excess return, etc.) as well as soft information conveyed in corporate 10-K disclosures such as positive disclosure tone (Buehlmaier & Whited, 2018; Tetlock et al., 2008). Since financial statement measures such as ROA and tone of 10-K communication can signal the size of potential financial gain if cyberattacks are successful, cyber criminals are likely to consider this before launching cyberattacks. With better financial prospects as well as 10-K filings presented in positive tone signifying higher potential financial gains, cyber criminals can easily rationalize their fraudulent behavior as worthwhile and worth the risk, thus, leading to a higher likelihood that resourceful and profitable companies as well as companies presenting 10-K communications in a positive tone will be attacked by cyber criminals.

Further, in their review of the evolution of fraud theory and its extensions, Dorminey, Fleming, Kranacher, and Riley (2012) discuss the recent recommendations "that the motivations of fraud perpetrators may be more appropriately expanded and identified with the acronym M.I.C.E (Kranacher et al., 2011)." (pp. 563). The expanded list of fraud theory motivations (i.e., M.I.C.E.) include money, ideology, coercion and ego/entitlement (Dorminey et al., 2012). We contend, based on fraud theory and its extended list of motivations, that most data breaches are likely carried out by cyber criminals who, motivated by the desire to get more money, are driven to launch cyberattacks on companies that have better financial prospects (e.g., resourceful companies and companies that communicate positive tone in their 10-K filings). Consequently, we are motivated to examine the effect that better financial prospects as well as 10-K disclosure tone have on the probability of data breaches.

Additionally, Hasan (2020) finds that less readable 10-K corporate communications may indicate less sophisticated management with poor governance practices/control systems. Since fraud theory indicates that weakness in systems presents opportunity for fraud commission, we are also motivated to examine the role that the readability of corporate 10-K communications plays in the likelihood of cyber criminals launching cyberattacks on companies. In addition to theoretical expositions motivating our study, the empirical evidence does provide some support for a possible association between data breaches and resourceful firms. For example, Florackis et al. (2023) find that indicators of resourcefulness such as higher cash reserve and size are positively related to their cybersecurity risk measure. Thus, both the (fraud) theoretical conjectures and empirical evidence support an association between data breaches and firm resourcefulness as well as tone and readability.

2.3. Relevant hypotheses: corporate communication and data breaches

Data breaches are generally motivated by the need to extort economic gains and corporate communication (e.g., 10-K reports) provides a richer information set about a firm's financial resources and future growth prospects. As Tetlock et al. (2008) point out,

textual analysis of corporate communication can capture hard-to-quantify components of a firm's fundamentals and provide useful information to market participants. Similarly, Buehlmaier and Whited (2018) note that, Management's Discussion and Analysis (MD&A) "contains a narrative explanation of the past performance of the firm, its financial condition, and its future prospects" (p. 2697). The need to decipher information from 10-K disclosures and evaluate the effect of that information on the likelihood of experiencing a data breach becomes even more paramount in recent years. Since 2011, SEC has mandated that publicly listed companies disclose their risk management measures and cybersecurity risks in their annual reports (Grant and Grant, 2014; SEC, 2007). Given the importance of narrative disclosure – as recognized by the SEC, investors and information intermediaries (e.g., credit rating agencies) – we are motivated to primarily examine the relation between the tone of narrative disclosure and the likelihood of data breach. Additionally, we focus on the impact of 10-K readability on the relation between 10-K tone and data breach likelihood.

2.3.1. 10-K disclosures and data breaches

Economic and financial prospects are generally the main opportunity that drive data breaches. For example, Verizon (2023) reports that 94.6% of data breaches in 2023 were financially motivated and Kamiya et al. (2021) find that targets of cyberattacks have higher future growth opportunities. While current financial statement data can be used to infer future growth prospects (e.g., Tobin's q as in Kamiya et al., 2021), we contend that textual analysis of mandatory 10-K disclosures is also an important gauge of future growth opportunities because not only does it provide insights on the quantifiable aspects of firm performance, but it also incorporates hard-to-quantify components of a firm's fundamentals (Dutta, Fuksa, & Macaulay, 2019; Tetlock et al., 2008). In fact, Li (2010) contends that the Management's Discussion and Analysis (MD&A) in 10-K disclosures is one of the most read and most important components of financial statements while Buehlmaier and Whited (2018) note that the MD&A "contains a narrative explanation of the past performance of the firm, its financial condition, and its future prospects" (p. 2697). As such, deciphering information on the future economic prospects of firms through textual analysis of 10-K disclosures could provide insights on the likelihood of experiencing cyberattacks. Understanding the role of 10-K disclosures in data breaches is even more necessary, especially given that since 2011, SEC has mandated that publicly listed companies disclose their risk management governance and cybersecurity risks in their annual reports (Grant and Grant, 2014; SEC, 2017).

Loughran and McDonald (2011) develop a dictionary for the analyses of financial text to decipher the sentiment and tone of text – as either positive, negative, litigious or constraining, which have been extensively used in the finance and accounting literature (e.g., Bodnaruk et al., 2015; Loughran & McDonald, 2016). We rely on Loughran and McDonald's dictionary in this study as well and obtain the relevant tones of 10-K disclosure directly from 'WRDS SEC Analytics Suite'. Our expectation is that when the tone of the disclosure is positive, then that would signal good performance and prospects for the company – which may make the company a potential target for cyberattacks. However, when the tone of the discussion is negative, litigious or constraining, that could signal less-than-stellar performance and prospects, which may make the company less attractive to fraudsters. Consequently, we hypothesize that:

Hypothesis 1. The likelihood of data breaches occurring would be high for firms with more positive sentiment in their annual report disclosures.

2.3.2. Readability of 10-K disclosures and data breaches

In crafting 10-K disclosures, management has the choice of making the disclosures simple and readable or complex and less readable. SEC (1998) calls for disclosures to be made in plain and easy-to-read language and yet, some companies choose to make disclosures in less readable forms. Less readable reports make it difficult for investors and information intermediaries (e.g., financial analysts) to gather insightful information about a firm. Using a measure of the readability of corporate 10-K filings, Lehavy et al. (2011) document that "analyst following, the amount of effort incurred to generate their reports, and the informativeness of their reports are greater for firms with less readable 10-Ks" (p. 1087). It implies that market participants have to invest in additional sources (e.g., greater analysts support) to quench their inquisitiveness about a firm – when it makes narrative disclosures less readable. Similarly, we argue that opacity of narrative disclosure is likely to increase the curiosity of hackers and data breachers in a firm and incentivize them to launch cyberattacks to retrieve valuable firm-specific information. Empirically, Florackis et al. (2023) find that firms with higher cybersecurity risk have 10-Ks that are less readable. Consequently, we expect that less readable disclosures will be positively related to the probability of data breaches. Therefore, we hypothesize that:

Hypothesis 2. The likelihood of data breaches occurring would be higher (lower) for firms that present their annual reports in less (more) readable form than firms that present their reports in more (less) readable forms.

Although examining the separate effects of tone and readability of 10-K corporate communications on the likelihood of successful data breaches are important, an examination of the interplay between these two factors may potentially provide a richer understanding of their joint impact on the likelihood of data breaches. Examining the interplay is particularly important since Tan, Wang, and Zhou (2014) argue for the exploration of the joint effects of readability and language tone because of their co-occurrence in practice. Indeed, Tan et al. (2014) find that language tone influences investors' earnings judgments when readability is low, but not when readability is high. Therefore, whether the interplay between tone and readability impact data breaches or not is worthy of exploration. In the context of our study, we expect that positive tone will have a positive association with data breach probability primarily in the firms with low narrative disclosure readability. Accordingly, we present the following hypothesis:

Hypothesis 3. In the firms with more readable (i.e., less complex) 10-K filings, positive tone is less likely to be associated with a higher data breach probability.

3. Data and methodology

3.1. Sample

We analyze data breaches from 2005 to 2018 using data collected by the Privacy Rights Clearninghouse (PRC). We first exclude breaches involving non-business cases, then manually match the remaining cases to firms listed in Compustat.[9] If we could not find a firm in Compustat, we additionally searched Bloomberg[10] and Google to ensure that issues with company names were not causing the mismatching with Compustat. After following these steps, our data breach sample consists of 491 observations (after matched with Compustat sample), made of 429 external cases and 62 internal cases.[11] To construct the sample for our regression analysis, we merge the data breach sample with several sources: the CRSP-Compustat merged database, readability and sentiment data from WRDS SEC Analytics Suite, and the BoardEx database. This data merging process is detailed in Panel A of Table 1. The requirement to align relevant data across these sources resulted in a sample size of 39,745 firm-year observations which includes 293 data breach observations and 5075 unique firms (similar to Kamiya et al., 2021).

3.2. Descriptive statistics and correlations

The descriptive statistics about our sample are presented in Table 1. Panel B provides descriptive statistics on the data breach dummy and firm characteristics. The data breach dummy has a mean of 0.00737– implying that 0.737% of the sample firms have experienced data breaches. Of particular importance are the descriptive statistics on annual report sentiment/tone and readability. The standard negative words ratio in 10-K disclosures is −2.5% while the standard positive words ratio is 1.4%. The average Fog Index of 20.11 is similar to the average in prior studies including the Fog Index of 19.7 reported by Kim, Wang, and Zhang (2019). This indicates that 10-K reports are, on average, very difficult to read, requiring almost 20 years of formal education (i.e., post-graduate education) to comprehend the typical 10-K disclosure on the first reading. Conclusions about the difficulty in reading 10-K disclosures are the same when we examine our alternative measure of readability, the Kincaid Index. Based on the Kincaid Index, 15.9 years of formal education is required to understand 10-K disclosures on the first read.

The bivariate Pearson correlations are presented in Table 2. As evident, the correlations between data breach variable, disclosure tone and readability variables are as expected. Specifically, the correlation between the positive words ratio and the data breach dummy variable is positive and significant. As well, both measures of readability (i.e., Fog Index and Kincaid Index) are significantly positively correlated with the data breach dummy – consistent with expectations. The correlation between the Fog Index and Kincaid Index (our measures of readability) is significantly high (0.968). This high correlation between the two measures of readability indicates that one should expect to get about the same results by using any one of the measures. Overall, the correlation between pairs of independent variables is generally low, signifying that multicollinearity is not an issue in this study. However, since bivariate correlations do not account for the effects of other variables, it is sometimes possible that the correlation coefficients are misleading and inconsistent with expectations (Golubov, Petmezas, & Travlos, 2012). Hence, it is essential to control for firm-specific characteristics to accurately reveal the relationship between firm disclosure and the probability of data breaches.

3.3. Baseline regression model of firm-year observation sample

Since we have a panel data structure with multiple observations per company and we are interested in understanding how differences in companies influence the likelihood of data breaches, we use the firm-level panel data random effects method[12] because Nickel (1981) highlights the superiority of random effects over fixed effects when one is interested in gleaning information through comparison of subjects.

We are primarily interested in the effect of disclosure tone on the likelihood of experiencing data breaches. The general form of our random-effects logit equation is:

$$\text{Likelihood of Data Breach} = \beta_0 + \beta_i\text{*Variable(s) of Interest} + \Sigma\, \beta_j\text{*Controls} + \text{Year fixed-effect} + \text{Industry fixed-effect} + \varepsilon$$

To analyze the effects of tone on the probability of being hacked, the key variable of interest is the standard positive words ratio; although we also include standard negative words ratio, standard uncertainty words ratio and standard litigious words ratio. We directly obtain these ratios, originally developed by Loughran and McDonald (2011), from 'WRDS SEC Analytics Suite' database.

[9] During this period, there were about 8127 cases reported. We excluded breaches involving nonbusiness (e.g., government and nongovernmental organizations) and focused on only business cases (i.e., PRC's organization type codes BSF - Businesses - Financial and Insurance Services, BSR - Businesses - Retail/Merchant and BSO - Businesses - Other), which narrowed the sample to about 2390 cases. Thereafter, we matched this sample manually with Compustat database identifier and arrive at a sample of 491 data breach cases.

[10] See http://www.bloomberg.com/Research/common/symbollookup/symbollookup.asp?lookuptype=public&region=ALL.

[11] We define internal data breaches as data breach types coded as INSD (insider breach) by the Privacy Rights Clearninghouse. We define external data breaches as those coded by the Privacy Rights Clearninghouse as CARD (payment card fraud), DISC (unintended disclosure), HACK (hacking or malware), PHYS (physical loss), PORT (portable device loss) and STAT (stationary device loss).

[12] Firm level panel data 'Random effect models' do not automatically account for year and industry fixed-effects. Hence, we have controlled for such fixed effects in our models.

Table 1
Sampling process and descriptive statistics
This table outlines the sampling process and summary statistics for our dataset. Panel A details the sampling process. We begin with selecting 491 unique data breach cases occurring from 2005 to 2018 following the process described in Section 3.1. In step 2, we merge the cases with the CRSP-Compustat merged database. In step 3, we aggregate these cases at the firm-year level, calculate the required firm-specific accounting variables, and eliminate observations with missing values, resulting in 53,709 unique firm-year observations. Step 4 involves incorporating self-disclosure and readability data. Finally, we merge the data with the governance index from BoardEx. Following the removal of missing values, we obtain a final dataset comprising 293 data breach cases and 39,745 firm-year level observations for firm-level analysis. Panel B presents descriptive statistics on the variables used in this study. All continuous variables are winsorized at the 1 and 99 percent level. The appendix presents a detailed description of each variable.

Panel A: Sampling process of the regression sample

Sampling process	# of data breaches	# of unique firms	# of firm-year observations
Step 1: Collect data breach cases	491	295	423
Step 2: Merge with CRSP-Compustat merged database	452	10,224	79,499
Step 3: Aggerage on firm-year level, drop missing accounting variables	336	7231	53,709
Step 4: Merge with WRDS self-disclosure and readability data	304	6045	44,361
Step 5: Merge with BoardEx database	293	5075	39,745

Panel B: Descriptive statistics

VARIABLES	(1)	(2)	(3)	(4)	(6)	(8)
	Obs	Mean	Std. Dev.	Min	Median	Max
Data Breach Dummy	39,745	0.007	0.086	0	0	1
Std. Positive Words Ratio	39,745	0.014	0.986	−4.229	−0.109	7.365
Std. Negative Words Ratio	39,745	−0.025	0.992	−4.354	−0.076	18.100
Std. Uncertainty Words Ratio	39,745	−0.050	0.960	−4.536	−0.043	4.117
Std. Litigious Words Ratio	39,745	0.000	0.996	−1.924	−0.280	11.280
Readability - Fog Index	39,745	20.110	1.194	15.420	20.030	93.890
Readability - Flesch-Kincaid Index	39,745	15.900	1.156	11.110	15.820	89.120
Book to Market Value	39,745	2.715	4.249	−12.800	1.875	30.510
Firm Size (Log (Total Assets))	39,745	6.665	2.052	−0.217	6.645	14.760
Leverage	39,745	0.168	0.192	0	0.105	0.893
ROA	39,745	−0.010	0.186	−1.217	0.024	0.285
R&D to Sales	39,745	0.179	0.962	0	0	9.384
Firm Age	39,745	21.060	16.230	1	16	92
Dividend Dummy	39,745	0.580	0.494	0	1	1
%Soft Assets	39,745	0.628	0.281	0.018	0.653	1.182
Abnormal Change in Employee Productivity	39,745	−0.145	0.819	−6.809	−0.026	1.909
Std. Deviation of ROA	39,745	0.091	0.490	0	0.033	51.230
Std. Deviation of Stock Return	39,745	0.652	1.859	0	0.376	133.800
Big-4 Auditor	39,745	0.713	0.453	0	1	1
Percent of Independent Board Members	39,745	0.705	0.137	0	0.714	0.917
Board Size	39,745	8.614	2.472	4	8	16
Cash to Total Assets	39,745	0.124	0.150	0.000	0.069	0.858
Excess Stock Return	39,745	0.052	0.501	−0.863	−0.022	2.470

Subsequently, we also examine the moderating effect of readability on the relationship between 10-K positive words ratio and data breach probability, by performing sub-group analysis. We use Fog Index and Kincaid Index as measures of readability. Appendix 2 presents a discussion on the readability index.

In our baseline regression models, we incorporate controls for other firm, and board characteristics. Our choice of control variables is motivated by the prior literature on data breaches and readability (e.g., Jiang et al., 2022; Kamiya et al., 2021; Lending et al., 2018; Li, 2008). We control for firm size as an indicator of resourcefulness (Kamiya et al., 2021), R&D to sales (Jiang et al., 2022) as an indicator for proprietary information and book-to-market, age, leverage (Jiang et al., 2022) and percent of soft assets (Dechow, Ge, Larson, & Sloan, 2011) – to capture firm complexity. Additionally, we include abnormal change in employee productivity to control for the likelihood of misstatement (Dechow et al., 2011). We include controls for dividend payments (Li, 2008), the standard deviation of ROA (Harford & Li, 2007), and the standard deviation of annual stock returns (Harford & Li, 2007) to account for financial performance stability. In terms of governance measures, we consider the percentage of independent board members for corporate governance quality (Beasley, 1996; Cheng, 2008; Falato, Kadyrzhanova, & Lel, 2014; Fama & Jensen, 1983) and board size for board composition and communication efficiency (Akbas, Meschke, & Wintoki, 2016; Cheng, 2008). Finally, we include proxies for high-quality auditors (Big Four auditors) to control for a firm's willingness to provide additional disclosures (Calderon & Gao, 2020; Hartmann & Carmenate, 2021). To mitigate the look ahead bias concern, the independent variables are measured one year before the data breach. See Appendix 1 for descriptions of the variables we use in this study. All variables used in this study are defined in Appendix 1.

Table 2
Correlation Matrix
This table presents the Pearson (pairwise) correlation coefficient between the variables used in the regression models. Definitions of variables are presented in the Appendix. Letters a, b, and c represent the significance at the 0.01, 0.05, and 0.10 levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) Data Breach Dummy	1.000											
(2) Std. Positive Words Ratio	0.019c	1.000										
(3) Std. Negative Words Ratio	0.001	0.038c	1.000									
(4) Std. Uncertainty Words Ratio	−0.011a	0.184c	0.289c	1.000								
(5) Std. Litigious Words Ratio	0.018c	−0.145c	0.274c	−0.480c	1.000							
(6) Readability - Fog Index	0.024c	0.026c	0.154c	0.035c	0.159c	1.000						
(7) Readability - Flesch-Kincaid Index	0.029c	0.089c	0.184c	0.078c	0.170c	0.968c	1.000					
(8) Book to Market Value	0.006	0.115c	−0.033c	0.011a	0.019c	0.015b	0.035c	1.000				
(9) Firm Size (Log (Total Assets))	0.126c	−0.046c	0.066c	−0.017c	0.139c	0.126c	0.123c	−0.030c	1.000			
(10) Leverage	0.016b	−0.078c	−0.068c	−0.112c	0.125c	0.042c	0.060c	−0.013a	0.239c	1.000		
(11) ROA	0.031c	−0.124c	−0.165c	−0.055c	−0.021c	−0.058c	−0.078c	0.015b	0.330c	−0.042c	1.000	
(12) R&D to Sales	−0.013b	0.228c	0.058c	0.058c	0.016b	0.049c	0.078c	0.064c	−0.190c	−0.025c	−0.486c	1.000
(13) Firm Age	0.046c	0.025c	−0.111c	−0.103c	0.072c	−0.026c	−0.022c	0.000	0.319c	0.053c	0.162c	−0.092c
(14) Dividend Dummy	0.047c	−0.077c	−0.121c	−0.054c	−0.018c	−0.010	−0.048c	−0.021c	0.428c	−0.021c	0.325c	−0.184c
(15) %Soft Assets	0.014b	−0.113c	0.053c	0.007	−0.059c	0.036c	−0.028c	−0.100c	0.231c	−0.097c	0.249c	−0.249c
(16) Abnormal Change in Employee Productivity	0.008	−0.143c	−0.054c	−0.031c	−0.045c	−0.051c	−0.071c	−0.032c	0.077c	0.004	0.175c	−0.273c
(17) Std. Deviation of ROA	−0.009	0.033c	0.040c	0.004	0.011a	0.011a	0.022c	0.037c	−0.128c	−0.015b	−0.139c	0.095c
(18) Std. Deviation of Stock Return	−0.005	0.021c	0.036c	0.024c	0.001	0.013b	0.024c	0.012a	−0.104c	−0.003	−0.062c	0.046c
(19) Big-4 Auditor	0.046c	0.156c	0.033c	−0.002	0.151c	0.077c	0.142c	0.078c	0.385c	0.180c	0.092c	0.017c
(20) Percent of Independent Board Members	0.042c	0.104c	0.045c	0.046c	0.062c	0.057c	0.074c	0.031c	0.302c	0.038c	0.064c	−0.014b
(21) Board Size	0.076c	0.011a	−0.002	−0.081c	0.081c	0.058c	0.030c	−0.014b	0.634c	0.080c	0.152c	−0.088c
(22) Cash to Total Assets	−0.019c	0.213c	0.110c	0.130c	−0.011a	0.039c	0.096c	0.174c	−0.383c	−0.215c	−0.278c	0.293c
(23) Excess Stock Return	0.002	0.013a	−0.032c	−0.020c	0.017c	−0.037c	−0.031c	0.154c	−0.021c	−0.013b	0.159c	−0.023c

	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
(13) Firm Age	1.000										
(14) Dividend Dummy	0.266c	1.000									
(15) %Soft Assets	−0.016b	0.183c	1.000								
(16) Abnormal Change in Employee Productivity	0.035c	0.089c	0.121c	1.000							
(17) Std. Deviation of ROA	−0.047c	−0.118c	−0.068c	−0.031c	1.000						
(18) Std. Deviation of Stock Return	−0.069c	−0.128c	−0.053c	−0.020c	0.574c	1.000					
(19) Big-4 Auditor	0.124c	0.093c	−0.147c	−0.025c	−0.016b	−0.023c	1.000				
(20) Percent of Independent Board Members	0.197c	0.137c	0.016b	−0.006	−0.038c	−0.038c	0.204c	1.000			
(21) Board Size	0.260c	0.336c	0.235c	0.045c	−0.066c	−0.079c	0.196c	0.249c	1.000		
(22) Cash to Total Assets	−0.142c	−0.231c	−0.423c	−0.156c	0.126c	0.089c	0.011a	−0.045c	−0.273c	1.000	
(23) Excess Stock Return	−0.015b	−0.071c	−0.001	0.024c	0.005	−0.007	0.036c	−0.021c	−0.034c	0.076c	1.000

4. Empirical results

4.1. Impact of disclosure sentiment on data breaches

We hypothesized that since mandatory 10-K disclosures contain difficult-to-quantify information about current firm performance and future growth prospects, textual extraction of the tone or sentiment conveyed in the disclosures would potentially impact the likelihood of experiencing a data breach. In particular, we conjecture that positive tone would increase the likelihood of being hacked since hackers may perceive such positive sentiment to imply significant financial heft and great future prospects. Table 3 presents the results that test this hypothesis.

As expected, the ratio of positive words extractable from 10-K disclosures has a significantly positive association with the probability of experiencing a data breach in column (1) (coefficient = 0.240, $p < 0.01$) – after controlling for other variables that might potentially affect the likelihood of being hacked. We also include three other measures of sentiment (negative words ratio, litigious words ratio and uncertainty words ratio) but we do not find a significant association between these sentiment variables and the chances of experiencing a data breach. This is not unexpected, however, since these sentiment measures do not necessarily convey opportunities that are exploitable by potential data breachers and may therefore not contribute to the chance of being breached. In column (2), we include governance measures (if the auditor is a Big 4 auditor, percent of independent board members, and board size) and the coefficient on the Std. Positive Words Ratio remains positive and significant (coefficient = 0.231, $p < 0.01$). This finding suggests that firms with better governance practices, as indicated by Big 4 auditor involvement and a higher proportion of independent directors, do not diminish the importance of the disclosure sentiment in financial communications. Overall, our results support hypothesis 1 that positive sentiment in firms' 10-K disclosures increases the likelihood of future data breaches. These results are also in line with the extended fraud theory as discussed in section 2.

4.2. Addressing endogeneity concerns

As discussed earlier, although our empirical settings mitigate some of the endogeneity concerns, our findings are still subject to common endogeneity issues inherent in corporate finance research. For instance, firms may adjust their narrative disclosures in anticipation of future cyberattacks (i.e., reverse causality), or certain characteristics like firm performance could influence both positive disclosures in 10-K filings and the probability of data breaches (omitted variable problem). To address these endogeneity issues, we employ various methods in this section including instrumental variable approach, matching firm analysis, and solely focusing on external data breaches.

4.2.1. Instrumental variable approach

First, we adopt an instrumental variable approach to mitigate endogeneity concerns both stemming from omitted variables and reverse causality issues. Following Ertugrul et al. (2017), we select the average positive words ratio of reports from firms located within the same area (sharing the first three digits of a 5-digit ZIP code) in a given year as our instrument. This instrumental variable meets the requirements of relevance and exclusion criteria: theoretically, firms in the same area tend to operate in similar environments and exhibit similar sentiments in their 10-K disclosures. As noted by Li, Wu, Xiao, and You (2019), the regional average sentiment should be strongly correlated with the sentiment of firms located within the same geographical location since the firm management tone shares a region-related component. Hence, the geographical average sentiment satisfies the relevance condition. On the other hand, it is unlikely that the average sentiment of disclosures in the same year and region would directly affect the individual firms' probability of data breaches; rather, data breach probability will be affected through firm-specific disclosure sentiments. This satisfies the principle of the exclusion criterion.[13] The use of the regional average measure as an instrument has been validated in previous studies (e.g., Dai, Fu, Kang, & Lee, 2016; Ertugrul et al., 2017; Li et al., 2019). The instrumental variable approach helps ensure that unobserved factors do not confound our estimations of the impact of disclosure sentiment on data breach probability.

In Table 4, we present the two-stage probit regression model using the regional mean of positive words ratio (Geo_mean_Positive_Words_Ratio) as an instrument. Column (1) presents the first-stage regression, which regresses the firm-level positive words ratio on Geo_mean_Positive_Words_Ratio and other control variables. We find that regional average positive words ratio (Geo_mean_Positive_Words_Ratio) is significant and positively associated with the firm-level positive words ratio (Std. Positive Words Ratio). This finding supports the relevance of our instrument. Furthermore, if the instrument is relevant but only weakly correlated with the endogenous variable (Std. Positive Words Ratio), the 2SLS estimator could be biased and unreliable (Bound, Jaeger, & Baker, 1995). We address this concern through weak instrument tests. The Cragg-Donald statistic estimated in the first stage exceeds the threshold of 23.1 suggested by Olea and Pflueger (2013). Furthermore, both Anderson-Rubin (AR) and Wald tests report significant statistics ($p < 0.01$), leading to the rejection of the null hypothesis that the instrumental variable is weak. By passing these tests, it confirms the adequacy and strength of the instrumental variable, justifying its use in the second stage of our analysis to estimate the true effect of disclosure sentiment on the probability of data breaches. In the second stage in column (2), we regress the data breach dummy on the instrumented positive words ratio and other control variables. The coefficient on the instrumented positive words ratio

[13] As noted by Roberts and Whited (2013), "Unlike the relevance condition, the exclusion condition cannot be tested because the regression error term, u, is unobservable" (p. 512). Thus, we follow the previous literature and rely on theoretical justifications and intuition to justify the exogeneity of the geographical average instrument (e.g., Dai et al., 2016; Ertugrul et al., 2017; Li et al., 2019).

Table 3
Relation between 10-K disclosure and probability of data breach
This table presents the logit regressions that show the impact of 10-K disclosure sentiment on the likelihood of experiencing a data breach incident (our dependent variable). Although our variable of interest is the positive words ratio (bolded), we also include three other measures of sentiment (negative words ratio, uncertainty words ratio and litigious words ratio) as well as other relevant control variables (e.g., firm size, leverage, R&D to sales, firm age, etc.) that may be associated with the likelihood of experiencing a data breach. In column (2), We also include controls for governance characteristics (if the auditor is a Big 4 auditor, percent of independent board members, and board size). All variable descriptions are presented in the Appendix. Regression models include industry dummies and year dummies. Test statistics are based on robust standard errors clustered at the firm level (in parentheses). ***, **, * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

VARIABLES	(1)	(2)
	Data Breach Dummy	Data Breach Dummy
Std. Positive Words Ratio	**0.240*** **	**0.231*** **
	(3.454)	**(3.276)**
Std. Negative Words Ratio	−0.025	−0.025
	(-0.294)	(-0.293)
Std. Uncertainty Words Ratio	−0.070	−0.078
	(-0.658)	(-0.733)
Std. Litigious Words Ratio	0.057	0.047
	(0.538)	(0.438)
Book to Market Value	−0.007	−0.007
	(-0.495)	(-0.521)
Firm Size (Log (Total Assets))	0.770***	0.761***
	(16.956)	(13.307)
Leverage	0.842*	0.837*
	(1.769)	(1.754)
ROA	4.653***	4.626***
	(3.776)	(3.738)
R&D to Sales	−0.052	−0.061
	(-0.141)	(-0.155)
Firm Age	−0.001	−0.002
	(-0.237)	(-0.322)
Dividend Dummy	0.152	0.147
	(0.738)	(0.714)
%Soft Assets	−0.319	−0.296
	(-0.891)	(-0.835)
Abnormal Change in Employee Productivity	0.008	0.009
	(0.066)	(0.073)
Std. Deviation of ROA	−0.057	−0.057
	(-0.963)	(-0.916)
Std. Deviation of Stock Return	0.032**	0.032**
	(2.270)	(2.290)
Big-4 Auditor		0.127
		(0.401)
Percent of Independent Board Members		0.720
		(1.224)
Board Size		−0.010
		(-0.275)
Constant	−14.137***	−14.535***
	(-15.426)	(-14.672)
Observations	39,745	39,745
Number of Unique Firms	5075	5075
Industry FE	YES	YES
Year FE	YES	YES

remains significant and positive ($p < 0.01$). Overall, this result supports our main finding in the paper that increased positivity in 10-K disclosure sentiment is associated with a higher probability of data breaches.

4.2.2. Matching firm analysis

In addition to the instrumental variable regression, we implement the propensity score matching (PSM) model to address the impact of potential confounding variables. Given that our data may not be randomly assigned to the treated and control groups, there is a possibility that firms with higher and lower positive words ratios may differ systematically across dimensions not captured by our control variables in the baseline regressions. Then the estimation of the treatment effect could be biased by the existence of other observed factors. To mitigate this issue, we create a more balanced matching sample using the propensity score matching as a robustness test to quantify the treatment effect (i.e., high vs. low positive words ratio) on firms' data breach probability (Rosenbaum & Rubin, 1983).

Specifically, we designate firm-year observations with a high positive 10-K disclosure sentiment (in the top decile of industry-year

Table 4
Instrumental Variable Analysis
This table reports the two-step instrumental probit regression at the firm-year level for data breach probability. Following Ertugrul et al. (2017), we select the average Std. Positive Words Ratio (Geo_mean_Positive_Words_Ratio) from firms located within the same area (sharing the first three digits of a 5-digit ZIP code) in a given year, as the instrumental variable. Columns (1) and (2) report regressions' first and second stages, respectively. The adequacy and strength of the instrumental variable are further evaluated through weak instrument tests of Cragg-Donald Wald F statistic, Anderson-Rubin (AR) and Wald tests. All models include industry and year fixed effects. ***, **, * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)
	Std. Positive Words Ratio	Data Breach Dummy
Geo_mean_Positive_Words_Ratio	**0.763*** **	
	(80.96)	
Instrumented (Std. Positive Words Ratio)		**0.257** **
		(3.05)
Std. Negative Words Ratio	0.0175**	−0.0229
	(3.28)	(-0.70)
Std. Uncertainty Words Ratio	0.0725***	−0.0335
	(12.24)	(-0.85)
Std. Litigious Words Ratio	−0.137***	0.0531
	(-24.48)	(1.40)
Book to Market Value	0.0121***	−0.00325
	(12.48)	(-0.50)
Firm Size (Log (Total Assets))	0.0183***	0.293***
	(5.47)	(15.21)
Leverage	−0.308***	0.347*
	(-13.04)	(2.22)
ROA	−0.112***	1.825***
	(-4.04)	(3.95)
R&D to Sales	0.105***	−0.0295
	(20.43)	(-0.11)
Firm Age	0.00105***	−0.000608
	(3.63)	(-0.40)
Dividend Dummy	−0.0332***	0.0646
	(-3.38)	(0.90)
%Soft Assets	−0.180***	−0.0697
	(-9.82)	(-0.57)
Abnormal Change in Employee Productivity	−0.0439***	0.0111
	(-8.23)	(0.23)
Std. Deviation of ROA	−0.00294	−0.0202
	(-0.29)	(-0.27)
Std. Deviation of Stock Return	0.00345	0.0119
	(1.29)	(1.04)
Big-4 Auditor	0.124***	−0.00495
	(11.38)	(-0.05)
Percent of Independent Board Members	0.342***	0.238
	(10.72)	(1.07)
Board Size	0.0299***	−0.0165
	(13.43)	(-1.17)
Constant	−0.320***	−5.790***
	(-9.32)	(-18.78)
Weak Instrument Tests:	Statistic	p-value
Cragg-Donald Wald F statistic	6554.505	
Anderson- Rubin (AR)	chi2(1) = 9.33	0.0023
Wald	chi2(1) = 9.32	0.0023
Observations	39,745	39,745
Number of Unique Firms	5075	5075
Industry FE	YES	YES
Year FE	YES	YES

Std. Positive Words Ratio) as the treated group, and those with lower positive sentiment as the control group (Std. Positive Words Ratio below industry-year median). We estimate a logit regression based on the control variables in our main regression (Table 3). We then select a matching control observation for each treated observation using the one-to-one nearest neighbor matching technique without replacement based on the estimated propensity scores (Sun, Dutta, Zhu, & Ren, 2021). With a caliper of 0.1, the matching process yields 3851 treatment and 3851 control firm-year observations, resulting in a final matched sample of 7702 firm-year observations.

The results of the matching process are presented in Table 5. To assess the quality of matching, we conduct difference-in-means tests of the matching variables between the treatment and control groups after matching in Panel A. The percentage biases for all covariates are reduced to below 10%, and the differences in mean values of all covariates are statistically insignificant after matching. This demonstrates that there is no significant difference between the treated and control groups regarding observed covariates after

Table 5
Matching firm analysis
In this table, we present the results of Propensity Score Matching. We matched the treatment sample (Std. Positive Words Ratio in the top industry-year decile) with a control sample (Std. Positive Words Ratio below the industry-year median) using the same set of controls as our baseline model in Table 3. We employ one-to-one matching with no replacement based on the propensity score generated by a logit regression. The matching yielded 3851 treated and 3851 control firm-year observations. Panel A displays the mean comparison of variables after matching between the two samples, while Panel B presents the average treatment effect of the treated on data breach probability between treated and control firms before and after matching.

Panel A

Variable	Mean			*t*-test	
	Treated	Control	%bias	t	p > t
Book to Market Value	3.193	3.269	−1.600	−0.690	0.487
Firm Size (Log (Total Assets))	6.979	7.039	−2.700	−1.170	0.242
Leverage	0.144	0.146	−1.100	−0.480	0.630
ROA	−0.015	−0.012	−1.200	−0.520	0.601
R&D to Sales	0.302	0.259	3.500	1.560	0.119
Firm Age	23.639	24.136	−2.600	−1.160	0.248
Dividend Dummy	0.620	0.627	−1.600	−0.680	0.495
%Soft Assets	0.598	0.600	−0.500	−0.240	0.812
Abnormal Change in Employee Productivity	−0.212	−0.224	1.300	0.550	0.579
Std. Deviation of ROA	0.082	0.083	−0.800	−0.360	0.718
Std. Deviation of Stock Return	0.608	0.629	−1.700	−0.730	0.464
Big-4 Auditor	0.818	0.816	0.700	0.290	0.768
Percent of Independent Board Members	0.727	0.729	−1.600	−0.720	0.470
Board Size	9.136	9.217	−3.100	−1.350	0.177

Panel B

	Treated	Controls	Difference	S.E.	t stat
Before match	0.014	0.006	0.008	0.001	5.310***
After match	0.014	0.008	0.006	0.002	**2.630*** **

matching, indicating an effective matching between the treatment and control firms.

Subsequently, Panel B displays the average treatment effect on the treated regarding the probability of a data breach for both treated and control firms before and after matching. The t-statistics remain significant at the 1% level (t-statistic = 2.63), indicating that a higher positive disclosure sentiment leads to an increased probability of data breaches. This finding corroborates our main finding in Table 3 and alleviates concerns that our results could be biased due to the observed covariates.

4.2.3. Analysis with only external data breaches

Since external data breaches are largely exogenous events and are not likely to be anticipated by the firm, we rerun our baseline regressions focusing solely on external data breaches. Specifically, we create a dummy variable that takes the value of 1 if the firm had an external data breach event and 0 otherwise.[14] We then regress the data breach dummy on our 10-K disclosure sentiment variable of interest (i.e., positive words ratio). The results are reported in Table 6, where the coefficient on the explanatory variable—positive sentiment expressed in annual reports—remains positive and significant (coefficient = 0.205, $p < 0.01$; coefficient = 0.2, $p < 0.01$). This finding alleviates concerns of reverse causality, specifically the concern that firms increase their positive words ratio in anticipation of future attacks.

4.3. Mechanisms through which disclosure sentiment affects the probability of data breaches

In this section, we explore the potential channels through which positive sentiment in 10-K reports increases the likelihood of data breaches. If the conjectures hold that companies with more positive disclosure sentiment are more attractive targets for data breaches due to the perceived economic gains from crimes committed against them, then it would follow that the relationship between positive sentiment in 10-K reports and the likelihood of data breaches would be stronger in firms exhibiting more resourcefulness and better financial prospects. These characteristics might include higher cash reserves or profitable stock returns, making them ideal targets for criminals seeking economic benefits through data breaches. To examine whether firms' more appealing characteristics moderate the relationship between the positive sentiment of disclosure and the likelihood of data breaches, we investigate the cross-sectional heterogeneity with different cash reserves and stock returns.[15] Specifically, we divide the sample into firms with high and low cash reserves groups (top and bottom tercile of Cash to Total Assets) and high and low stock returns groups (top and bottom tercile of Excess

[14] We define external data breaches as those coded by the Privacy Rights Clearninghouse as CARD (payment card fraud), DISC (unintended disclosure), HACK (hacking or malware), PHYS (physical loss), PORT (portable device loss) and STAT (stationary device loss).

[15] We include only the positive word ratio for brevity, noting that the results remain the same when other sentiment measures are included.

Table 6
Relation between 10-K disclosure and probability of external data breach
Table 6 presents the logit regressions that show the impact of 10-K disclosure sentiment on the likelihood of experiencing an external data breach incident (our dependent variable). To address endogeneity concerns, we exclude internal data breaches and run the regressions using external data breaches only. We regress the external data breach dummy on our 10-K disclosure sentiment variable of interest (i.e., positive words ratio), the three other measures of sentiment (negative words ratio, uncertainty words ratio and litigious words ratio) and other relevant control variables (e.g., firm size, leverage, R&D to sales, firm age, etc.) that may have an impact on the likelihood of experiencing a data breach. In column (2), we also include controls for governance characteristics (if the auditor is a Big 4 auditor, percent of independent board members, and board size). All variable descriptions are provided in the Appendix. Regression models include industry dummies and year dummies. Test statistics are based on robust standard errors clustered at the firm level (in parentheses). ***, **, * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

VARIABLES	(1)	(2)
	External Data Breach	External Data Breach
Std. Positive Words Ratio	**0.205*** **	**0.200*** **
	(2.796)	**(2.693)**
Std. Negative Words Ratio	−0.044	−0.046
	(-0.477)	(-0.500)
Std. Uncertainty Words Ratio	−0.054	−0.061
	(-0.499)	(-0.562)
Std. Litigious Words Ratio	0.051	0.044
	(0.455)	(0.396)
Book to Market Value	−0.004	−0.004
	(-0.303)	(-0.312)
Firm Size (Log (Total Assets))	0.747***	0.739***
	(15.887)	(12.571)
Leverage	0.810	0.782
	(1.548)	(1.492)
ROA	4.087***	4.024***
	(3.440)	(3.387)
R&D to Sales	−0.162	−0.183
	(-0.273)	(-0.285)
Firm Age	0.000	0.000
	(0.009)	(0.022)
Dividend Dummy	0.174	0.175
	(0.800)	(0.811)
%Soft Assets	−0.082	−0.055
	(-0.223)	(-0.151)
Abnormal Change in Employee Productivity	−0.021	−0.020
	(-0.164)	(-0.163)
Std. Deviation of ROA	−0.058	−0.057
	(-0.871)	(-0.845)
Std. Deviation of Stock Return	0.031**	0.031**
	(2.005)	(1.990)
Big-4 Auditor		0.249
		(0.739)
Percent of Independent Board Members		0.251
		(0.435)
Board Size		−0.012
		(-0.324)
Constant	−14.442***	−14.632***
	(-14.413)	(-13.401)
Observations	39,745	39,745
Number of Unique Firms	5075	5075
Industry FE	YES	YES
Year FE	YES	YES

Stock Return). We perform our main regression to examine the impact of positive tone (Std. Positive Words Ratio) in the subsamples separately.[16]

Columns (1) and (2) of Table 7 present the results from subsample regressions where firms are grouped by high or low cash reserves. In column (1), the coefficient of positive words ratio is positive and significant among firms with high levels of cash reserves (coefficient = 0.386, $p < 0.05$). Conversely, this coefficient is insignificant for firms with low levels of cash reserves in column (2)

[16] Since, we use a logistic regression (where the dependent variable is dichotomous), we avoid interpreting our results using an interaction term and instead use sub-sample analysis. While interpretation of an interaction term is straightforward in a linear model (e.g. OLS), one has to be cautious with interpretations in a nonlinear model (e.g. Logit, Probit). The interaction effect "cannot be evaluated simply by looking at the sign, magnitude, or statistical significance of the coefficient on the interaction term when the model is nonlinear. Instead, the interaction effect requires computing the cross derivative or cross difference" (Ai & Norton, 2003, p. 129).

Table 7
Potential Mechanisms
This table explores the potential channels through which positive sentiment in 10-K reports increases the likelihood of data breaches with sub-sample analysis. The dependent variable is the data breach dummy. Columns (1) and (2) present the logit regressions of subsamples classified by cash reserves (Cash to Total Assets). Columns (3) and (4) present the logit regressions of subsamples classified by stock returns (Excess Stock Return). We split our sample into high and low cash reserves subgroups (top and bottom tercile of Cash to Total Assets) and high and low stock returns subgroups (top and bottom tercile of Excess Stock Return). Within these subgroups, we regress the data breach dummy on our 10-K disclosure sentiment variable of interest (i.e., positive words ratio) and other relevant control variables (e.g., firm size, leverage, R&D to sales, firm age, etc.) that may have an impact on the likelihood of experiencing a data breach. All variable descriptions are provided in the Appendix. Regression models include industry dummies and year dummies. Test statistics are based on robust standard errors clustered at the firm level (in parentheses). ***, **, * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

Subgroups	(1)	(2)	(3)	(4)
	Top Tercile	Bottom Tercile	Top Tercile	Bottom Tercile
	High Cash to Total Assets	Low Cash to Total Assets	High Excess Stock Return	Low Excess Stock Return
Std. Positive Words Ratio	**0.386****	**0.200**	**0.496*****	**0.123**
	(2.498)	**(1.625)**	**(3.853)**	**(0.847)**
Book to Market Value	−0.023	−0.017	−0.011	−0.022
	(-0.910)	(-0.726)	(-0.449)	(-0.652)
Firm Size (Log (Total Assets))	0.576***	0.995***	0.754***	0.661***
	(6.585)	(10.350)	(8.839)	(6.780)
Leverage	1.241**	1.417	−0.143	0.639
	(2.035)	(1.496)	(-0.204)	(0.747)
ROA	3.176**	3.508	4.937**	6.897***
	(2.138)	(1.423)	(2.118)	(3.550)
R&D to Sales	−0.528	0.109	0.138	0.531*
	(-0.491)	(0.454)	(0.480)	(1.809)
Firm Age	−0.004	0.001	−0.003	0.001
	(-0.419)	(0.117)	(-0.417)	(0.085)
Dividend Dummy	0.108	−0.006	0.042	−0.248
	(0.345)	(-0.016)	(0.140)	(-0.684)
%Soft Assets	−0.675	0.717	−0.534	0.126
	(-1.289)	(1.074)	(-0.978)	(0.208)
Abnormal Change in Employee Productivity	0.098	0.025	−0.015	0.357
	(0.610)	(0.106)	(-0.073)	(0.862)
Std. Deviation of ROA	−0.599	−0.058	−1.825	−0.831
	(-0.562)	(-0.077)	(-1.266)	(-0.551)
Std. Deviation of Stock Return	0.116***	0.085	0.077**	0.087*
	(2.727)	(1.326)	(2.296)	(1.707)
Big-4 Auditor	0.648	−0.599	−0.056	0.799
	(1.146)	(-1.197)	(-0.116)	(1.202)
Percent of Independent Board Members	0.838	0.555	0.480	1.615
	(0.904)	(0.566)	(0.519)	(1.493)
Board Size	0.050	−0.136**	−0.009	0.062
	(0.867)	(-1.970)	(-0.147)	(0.983)
Constant	−11.795***	−15.292***	−13.648***	−15.339***
	(-8.118)	(-8.165)	(-7.391)	(-9.011)
Observations	11,539	12,511	13,191	11,538
Number of Unique Firms	2812	2846	4210	3848
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES

(coefficient = 0.2, $p > 0.1$). Columns (3) and (4) of Table 7 display the results from subsample regressions where firms are grouped by high or low stock returns. In column (3), the coefficient of positive words ratio is positive and significant among firms with better stock performance (coefficient = 0.496, $p < 0.01$). However, in column (4), the coefficient for positive words ratio is not statistically significant for firms with low excess stock returns (coefficient = 0.123, $p > 0.1$). The results suggest that the positive effect of the positive tone on the likelihood of data breaches is dominant in firms with greater cash reserves and stronger stock performance. This indicates that the firms' resourcefulness and promising future prospects, which make them more appealing to potential hackers, serve as potential channels through which positive sentiment in firm disclosures correlates with a higher probability of data breaches.

Consistent with the predictions of fraud theory (Cressey, 1950; Wolfe & Hermanson, 2004), these findings suggest that cyber criminals may target companies with promising financial outlooks and high stock profitability. The rationale behind these attacks is that the potential financial rewards from a successful cyberattack outweigh the associated risks. This belief leads to an increased likelihood that such firms will face data breaches, as the benefits are deemed to justify the potential consequences.

4.4. Disclosure readability, disclosure sentiment and data breaches

In this section, we explore the impact of the readability of firm disclosures on data breaches and examine how readability influences

the relationship between disclosure sentiment and the likelihood of data breaches. Columns (1) and (2) in Table 8 present the results of the association between annual report readability and the chances of suffering a data breach. We operationalize annual report readability using the Fog Index and Kincaid index. In column (1), the significant positive association (at the 1% level) between the Fog Index and likelihood of a data breach signifies that, as the Fog Index increases (which represents less readable text), the chances of experiencing a data breach also increase.

The same conclusions are arrived at when we examine the association between the Kincaid Index, our alternative measure of annual report readability, and the likelihood of being hacked. Higher values of the Kincaid Index (which signify difficult-to-read documents) increase the likelihood of experiencing a data breach. The coefficient of the Kincaid Index is positive and significant at the 1% in column (2). These results provide support for our hypothesis 2 that the likelihood of data breach is higher (lower) for the firms that present their annual reports in poorer (better) readable form.

Flagging the co-occurrence between language sentiment and readability, Tan et al. (2014) highlight the need to explore the joint effects of readability and language tone. We investigate this interplay by grouping firms into terciles based on readability. The top tercile represents the low readability group while the bottom tercile represents the high readability group. Consistent with Tan et al. (2014), we find different effects between the top tercile (low readability group) and bottom tercile (high readability group). Specifically, we find that positive tone (positive words ratio) is significantly positively related to the chances of experiencing data breaches only in situations where annual reports are less readable (column (3) with high Fog Index and column (5) with high Flesch-Kincaid Index in Table 8). For the high readability group, positive tone leads to lower but statistically insignificant chances of data breaches (columns (4) and (6) in Table 8). These findings show the primacy of corporate disclosure clarity in alleviating the chances of data breaches. These results support our hypothesis 3 that in the firms with more readable (i.e., less complex) 10-K filings, positive tone is less likely to be associated with a higher data breach probability.

In summary, our results indicate that disclosure sentiment, readability and their interactions have significant ability to predict the likelihood of being breached. These results support that the tone and readability of corporate communication (annual reports) will be key determinants in predicting the likelihood of a data breach.

5. Additional tests

5.1. Economic effects of data breaches

After the evaluation of our main research questions, we further verify whether the sampled data breaches have significant economic effects on shareholders. To this effect, we examine the cumulative abnormal returns (CARs) over various event windows from a three-day window (−1 to +1) surrounding the announcement of a data breach to an eleven-day window (−5 to +5) surrounding the announcement of a data breach. For all of the data breaches,[17] the CARs for the four event windows that we evaluate are significantly negative at the 1% level – signifying a significantly negative stock price reaction on the announcement of data breaches (Table 9). The CARs range from −0.648% (over the [−1,+1] window) to −1.035% over the [−2,+2] window (Panel A, Table 9). When we partition our sample into external data breaches and internal data breaches, we find that for the external data breaches, the CARs are negative and significant (at the 1% level) for all the four event windows that we examine. In particular, the CAR of −0.699% over the three-day window surrounding the announcement of data breaches (i.e., [−1,+1] window) is similar to the CAR of −0.84% that Kamiya et al. (2021) find over the same window for their study of external data hacks (Panel B, Table 9). For the internal data breaches, the CAR is significantly negative (at the 5% level) for the [−2,+2] window and negative but marginally significant at the 10% level for the [−3, +1] window. The other two event windows (i.e., [−1,+1] and [−5,+5]) for internal data breaches have CARs that are negative but statistically indistinguishable from zero (Panel C, Table 9).

Overall, the sampled data breaches are significant economic events that result in significant shareholder wealth destruction once they are publicly announced. This finding reiterates the importance of examining various factors (e.g., the readability of narrative disclosure) that may affect the probability of data breach.

5.2. Rare event logit regression

The occurrence of a data breach is a rare event (with a mean value of 0.00737), as the majority of firm-year observations have recorded no data breaches. This could pose challenges for our logistic regression estimation since logit regression might be biased when the events per variable (EPV) ratio is too small (Vittinghoff & McCulloch, 2007). To alleviate the concern about the imbalanced sample, we adopt the rare events logistic regression technique developed by King and Zeng (2001) [18].

[17] We utilize all available 491 data breach cases that can be merged with the CRSP-Compustat merged database to obtain available stock returns, as outlined in step 2 of Panel A. Since we require current return data but not lagged variables in the regression sample, one case is lost compared to the sample in step 2 (451 vs. 452).

[18] Constructing a matched sample would be ideal in the analysis. However, traditional matching firm analyses such as Propensity Score Matching (PSM) are usually used by matching between treated and control groups (in our case, the 10-K report positive tones) to control for confounding variables that might affect the treatment assignment, making it inappropriate to solve the imbalance that exists in the dependent variable (in our case, data breaches). Thus, we turn to King and Zeng (2001)'s correction method, which adjusts the logistic regression estimates to account for the small sample bias

Table 8
The interplay between 10-K disclosure sentiment and readability on the likelihood of data breach
This table presents the logit regressions that show the effect that the interplay between 10-K disclosure sentiment and readability has on the likelihood of experiencing a data breach. The dependent variable is the data breach dummy. Columns (1) and (2) present the logit regressions that show the impact of 10-K disclosure readability (Fog Index and Kincaid Index) on the likelihood of experiencing a data breach based on the full sample. Columns (3)–(6) present the logit regressions that show the effect that the interplay between 10-K disclosure sentiment and readability has on the likelihood of experiencing a data breach incident (our dependent variable) with sub-sample analysis. We first group our sample into low readability disclosures (top tercile of Fog Index and Kincaid Index) and high readability disclosures (bottom tercile of Fog Index and Kincaid Index). Within these readability groups, we then regress the data breach dummy on our 10-K disclosure sentiment variable of interest (i.e., positive words ratio), the three other measures of sentiment (negative words ratio, uncertainty words ratio and litigious words ratio) and other relevant control variables (e.g., firm size, leverage, R&D to sales, firm age, etc.) that may have an impact on the likelihood of experiencing a data breach. All variable descriptions are provided in the Appendix. Regression models include industry dummies and year dummies. Test statistics are based on robust standard errors clustered at the firm level (in parentheses). ***, **, * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

Subgroups	(1)	(2)	(3)	(4)	(5)	(6)
	Full sample		Top Tercile	Bottom Tercile	Top Tercile	Bottom Tercile
			High Fog Index (Low Readability)	Low Fog Index (High Readability)	High Flesch-Kincaid Index (Low Readability)	Low Flesch-Kincaid Index (High Readability)
Readability - Fog Index	**0.090***** **(3.457)**					
Readability - Flesch-Kincaid Index		**0.091***** **(3.580)**				
Std. Positive Words Ratio			**0.322***** **(3.522)**	**−0.119** **(-0.770)**	**0.275***** **(2.904)**	**−0.226** **(-1.259)**
Std. Negative Words Ratio			−0.020 (-0.151)	0.018 (0.099)	0.050 (0.424)	0.234 (0.963)
Std. Uncertainty Words Ratio			−0.238 (-1.470)	−0.090 (-0.473)	−0.313** (-2.238)	−0.406 (-1.595)
Std. Litigious Words Ratio			−0.004 (-0.030)	−0.103 (-0.407)	−0.031 (-0.237)	−0.405* (-1.788)
Book to Market Value	−0.005 (-0.346)	−0.005 (-0.349)	−0.015 (-0.555)	−0.045** (-2.112)	−0.010 (-0.359)	−0.031 (-1.354)
Firm Size (Log (Total Assets))	0.750*** (13.230)	0.747*** (13.211)	0.792*** (9.489)	0.744*** (6.426)	0.806*** (9.463)	0.758*** (5.693)
Leverage	0.810* (1.675)	0.807* (1.674)	0.816 (1.327)	0.708 (0.628)	0.473 (0.664)	0.962 (0.832)
ROA	4.990*** (3.950)	4.976*** (3.940)	6.248*** (2.927)	2.163 (1.498)	5.207** (2.541)	4.302** (2.079)
R&D to Sales	−0.004 (-0.011)	−0.021 (-0.055)	−0.147 (-0.259)	−0.125 (-0.207)	−0.271 (-0.267)	−0.572 (-0.186)
Firm Age	−0.000 (-0.007)	0.000 (0.012)	0.005 (0.591)	−0.005 (-0.601)	−0.000 (-0.034)	0.011 (1.140)
Dividend Dummy	0.159 (0.790)	0.162 (0.804)	−0.172 (-0.590)	0.430 (1.039)	−0.124 (-0.443)	0.309 (0.651)
%Soft Assets	−0.327 (-0.940)	−0.316 (-0.908)	−0.704 (-1.340)	−0.007 (-0.010)	−0.612 (-1.196)	0.931 (1.220)
Abnormal Change in Employee Productivity	0.007 (0.052)	0.007 (0.054)	−0.059 (-0.357)	0.359 (0.987)	−0.156 (-1.201)	0.382 (0.827)
Std. Deviation of ROA	−0.067 (-0.885)	−0.067 (-0.876)	0.047 (0.238)	0.291*** (3.099)	−1.340 (-0.992)	0.253** (2.309)
Std. Deviation of Stock Return	0.032** (2.329)	0.032** (2.329)	−0.013 (-0.217)	−0.002 (-0.018)	−0.036 (-0.606)	0.159*** (4.024)
Big-4 Auditor	0.154 (0.486)	0.150 (0.473)	−0.224 (-0.496)	0.468 (0.652)	−0.045 (-0.096)	−0.428 (-0.666)
Percent of Independent Board Members	0.906 (1.534)	0.898 (1.521)	−0.435 (-0.572)	0.114 (0.097)	0.019 (0.024)	1.425 (0.995)
Board Size	0.001 (0.040)	0.001 (0.040)	−0.003 (-0.053)	0.032 (0.444)	−0.040 (-0.747)	0.034 (0.394)
Constant	−16.340*** (-14.493)	−15.965*** (-14.991)	−14.161*** (-10.349)	−14.755*** (-6.588)	−13.130*** (-10.347)	−14.108*** (-8.078)
Observations	39,745	39,745	12,421	12,511	12,421	9219
Number of Unique Firms	5075	5075	3037	2704	3019	2344
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES

King and Zeng (2001) introduce a modified logistic regression method specifically designed for rare events data, which they call ReLogit. This approach corrects the biases that occur in conventional logistic regression when dealing with datasets that have a severe imbalance between the rare event and non-events. The ReLogit model assigns weights to events and non-events to correct for the

Table 9
Impact of data breach events - Univariate CAR Analysis
This table presents the cumulative abnormal returns (CARs) for a number of windows surrounding the public announcement of data breaches. Panel A presents CARs for all data breaches, Panel B reports CARs for external data breaches while Panel C presents CARs for internal data breaches. We define internal data breaches as data breach types coded as INSD (insider breach) by the Privacy Rights Clearninghouse. We define external data breaches as those coded by the Privacy Rights Clearninghouse as CARD (payment card fraud), DISC (unintended disclosure), HACK (hacking or malware), PHYS (physical loss), PORT (portable device loss) and STAT (stationary device loss). ***, **, * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A. All Data Breach Cases (Both External/Internal)				
	# observations	Mean (%)	t-stat	p-value
CAR (−1 to +1)	451	−0.648	−4.628***	0.000
CAR (−2 to +2)	451	−1.035	−5.293***	0.000
CAR (−3 to +1)	451	−0.971	−5.665***	0.000
CAR (−5 to +5)	451	−0.967	−3.552***	0.000
Panel B. All External Data Breach Cases				
	# observations	Mean (%)	t-stat	p-value
CAR (−1 to +1)	389	−0.699	−4.480***	0.000
CAR (−2 to +2)	389	−1.048	−4.843***	0.000
CAR (−3 to +1)	389	−0.975	−5.334***	0.000
CAR (−5 to +5)	389	−1.007	−3.382***	0.001
Panel C. All Internal Data Breach Cases				
	# observations	Mean (%)	t-stat	p-value
CAR (−1 to +1)	62	−0.327	−1.172	0.246
CAR (−2 to +2)	62	−0.953	−2.224**	0.030
CAR (−3 to +1)	62	−0.951	−1.921*	0.059
CAR (−5 to +5)	62	−0.675	−1.084	0.283

sampling design's inherent bias, ensuring that the estimation process is consistent and efficient. This weighting scheme provides unbiased and accurate estimates of model parameters, which are crucial for predictive accuracy in rare events data. This correction addresses the influence of each group on the parameter estimates, mitigating issues of overrepresentation or underrepresentation due to the sampling design.

The results of the rare event regression are reported in Table 10. After for adjusting the weights of the rare data breach events, the coefficients on Std. Positive Words Ratio remains positive and significant, with and without the controls for governance measures (coefficient = 0.212, $p < 0.01$; coefficient = 0.202, $p < 0.01$). The robustness of our results aligns with the underlying assumption in the literature on rare event logistic regression, which posits that the absolute frequency of the rarer event is critical, not its proportion in the dataset (Vittinghoff & McCulloch, 2007). Further, "bias can be reduced by increasing the total sample size while keeping EPV constant (i.e., increasing the number of non-events)" (Van Smeden et al., 2016, p. 8). In our study, the presence of approximately 300 data breach cases and a total sample size of 39,745 firm year observations, while not a larger sample size, is nonetheless adequate and offers a reasonably good basis for insightful analysis of such rare events.

6. Summary and conclusions

Motivated by the need to understand the impact that corporate communication, specifically tone/sentiment and readability of annual reports, have on the odds of experiencing a data breach, this paper provides the first evidence on the effects of sentiment and readability of annual reports on the likelihood of being breached. We find that positive sentiment expressed in 10-K disclosures significantly increases the odds of data breaches. We also find that 10-K reports presented in difficult to read language (low readability) increase the odds of experiencing data breaches, and that when readability is interacted with sentiment, the positive sentiment is only significantly related to the odds of experiencing data breaches for firms in the low readability group. These results indicate that corporate communication tone and transparency could affect the unwarranted inquisitiveness of data breachers and hackers.

To ensure the robustness of our results we adopted several approaches. As a standard procedure, we control for industry and year fixed effects and used a comprehensive set of control variables in the analysis. To address the issue of a set of smaller data breach events in our sample, we applied rare event analysis and confirmed the robustness of results. More importantly, we have performed various tests including, (a) instrumental variable analysis, (b) matching firm analysis, and (c) focusing only on external data breaches – to alleviate the concerns of endogeneity to a larger extent. However, we admit that like in most of the corporate finance research that rely on archival data, it is extremely challenging to fully address the issue of endogeneity bias.

Subsequently, we explore the related mechanisms find strong evidence that the association between positive tone of annual reports and odds of a data breach is largely attributable to a firm's internal resources and market prospects. These findings are consistent with the predictions of fraud theory (Cressey, 1950; Wolfe & Hermanson, 2004). Given that companies with strong financial performance and promising prospects could present lucrative opportunities for cyberattacks, cyber criminals are often motivated to initiate attacks on such firms. They believe that the financial gains of a successful cyberattack outweigh the associated risks, thereby heightening the

Table 10
Rare event logit regression
Table 10 presents the logit regressions that show the impact of 10-K disclosure sentiment on the likelihood of experiencing a data breach incident using rare logit regression (ReLogit) proposed by King and Zeng (2001). The modified logistic regression technique ReLogit corrects the biases inherent in conventional logistic regression analyses that arise when dealing with datasets with a severe imbalance between rare events and non-events. We regress the data breach dummy on our 10-K disclosure sentiment variable of interest (i.e., positive words ratio), the three other measures of sentiment (negative words ratio, uncertainty words ratio and litigious words ratio) and other relevant control variables (e.g., firm size, leverage, R&D to sales, firm age, etc.) that may have an impact on the likelihood of experiencing a data breach. In column (2), We also include controls for governance characteristics (if the auditor is a Big 4 auditor, percent of independent board members, and board size). All variable descriptions are provided in the Appendix. Regression models include industry dummies and year dummies. Test statistics are based on robust standard errors clustered at the firm level. ***, **, * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

VARIABLES	(1)	(2)
	Data Breach Dummy	Data Breach Dummy
Std. Positive Words Ratio	**0.212*** **	**0.202*** **
	(3.195)	**(3.004)**
Std. Negative Words Ratio	−0.051	−0.053
	(-0.622)	(-0.643)
Std. Uncertainty Words Ratio	−0.050	−0.062
	(-0.516)	(-0.632)
Std. Litigious Words Ratio	0.096	0.081
	(0.934)	(0.778)
Book to Market Value	−0.001	−0.002
	(-0.084)	(-0.123)
Firm Size (Log (Total Assets))	0.715***	0.719***
	(15.878)	(12.262)
Leverage	0.804	0.786
	(1.591)	(1.553)
ROA	4.576***	4.550***
	(3.588)	(3.545)
R&D to Sales	1.399*	1.344
	(1.663)	(1.532)
Firm Age	−0.002	−0.002
	(-0.389)	(-0.343)
Dividend Dummy	0.179	0.183
	(0.860)	(0.888)
%Soft Assets	−0.139	−0.111
	(-0.379)	(-0.305)
Abnormal Change in Employee Productivity	−0.005	−0.005
	(-0.038)	(-0.039)
Std. Deviation of ROA	0.552***	0.599***
	(11.112)	(11.467)
Std. Deviation of Stock Return	0.051***	0.052***
	(4.309)	(4.352)
Big-4 Auditor		0.184
		(0.578)
Percent of Independent Board Members		0.706
		(1.235)
Board Size		−0.032
		(-0.866)
Constant	−11.239***	−11.660***
	(-19.071)	(-17.465)
Observations	39,745	39,745
Number of Unique Firms	5075	5075
Industry FE	YES	YES
Year FE	YES	YES

risk of data breaches for these companies.

Our findings offer practical insights for organizations seeking to enhance their cybersecurity posture. The observed relationships between positive sentiment, readability, and data breach likelihood provide support the implementation of communication transparency initiatives as a proactive risk mitigation strategy. Based on our findings, we contend that firms expressing more positive sentiment in their 10-K disclosures are more susceptible to data breaches, underscoring the need for enhanced data security measures in these organizations. Additionally, firms aiming to curb the rising tide of data breaches may benefit from improving the readability of their corporate communications. This is particularly important for firms displaying attractive characteristics, such as substantial cash holdings or profitable stock returns, which may make them more appealing targets for cyberattacks. Furthermore, our confirmation and extension of prior empirical findings, such as the negative market reaction to data breach announcements and the moderating effects of firm characteristics on sentiment-breaches relationship, underscore the robustness and generalizability of these phenomena

across different contexts. This strengthens the empirical foundation for policy recommendations and managerial interventions aimed at bolstering cybersecurity resilience.

Although sentiment and readability are important aspects of firm communications, our findings may be subject to limitations inherent in the measures employed. While our study utilizes textual analysis techniques to gauge sentiment and readability, it does not explore more nuanced text measures. Future research could explore additional dimensions of textual disclosure, such as changes in disclosure tones (Cohen, Malloy, & Nguyen, 2020). This could provide further insights into how subtle shifts in tone might influence perceptions and potentially impact the likelihood of cyberattacks. Additionally, future studies could also explore the role of industry-specific factors and regulatory environments in shaping the relationship between communication transparency and cybersecurity outcomes and could offer valuable insights into sector-specific vulnerabilities and risk management strategies. Moreover, the emergence of novel communication channels, such as social media and non-traditional reporting platforms, presents opportunities for exploring their impact on cybersecurity risk perception and response strategies.

Author statement

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Appendix 1. Variable Descriptions

Variable Name	Variable Description	Source
Dependent Variables		
Data Breach Dummy	Dummy variable that takes the value of 1 if the firm had a data breach event and 0 otherwise (Kamiya et al., 2021).	PRC
Independent Variables		
Std. Positive Words ratio	Percentage of positive words in the 10-K filing. The list consists of 353 words including variations. The positive list contains words such as achieve, attain, efficient, improve, profitable, and upturn. The ratio is standardized by subtracting the mean value and dividing by the standard deviation (Loughran & McDonald, 2011).	WRDS SEC Analytics Suite
Std. Negative Words ratio	Percentage of negative words in the 10-K filing. The list includes 2337 words, such as loss, adverse, discontinued, penalties, unpaid, investigation, misstatement, misconduct, forfeiture, serious, allegedly, noncompliance, deterioration, and felony. The ratio is standardized by subtracting the mean value and dividing by the standard deviation (Loughran & McDonald, 2011).	WRDS SEC Analytics Suite
Std. Uncertainty Words Ratio	Percentage of uncertain words in the 10-K filing. The uncertainty list includes words denoting uncertainty, with emphasis on the general notion of imprecision rather than exclusively focusing on risk. The list includes 285 words, such as approximate, contingency, depend, fluctuate, indefinite, uncertain, and variability. The ratio is standardized by subtracting the mean value and dividing by the standard deviation (Loughran & McDonald, 2011).	WRDS SEC Analytics Suite
Std. Litigious Words Ratio	Percentage of litigious words in the 10-K filing. The list categorizes words reflecting a propensity for legal contest or, per our label, litigiousness. The list includes 731 words such as claimant, deposition, interlocutory, testimony, and tort. The ratio is standardized by subtracting the mean value and dividing by the standard deviation (Loughran & McDonald, 2011).	WRDS SEC Analytics Suite
Readability - Fog Index	Fog Index is a readability measure that combines the number of words per sentence and the percent of complex words to determine the number of years of formal education required to read text and understand it the first time (Li, 2008).	WRDS SEC Analytics Suite
Readability - Flesch-Kincaid Index	Kincaid Index is a readability measure that combines the number of words per sentence and the number of syllables per word to determine the number of years of formal education required to read text and understand it the first time (Li, 2008).	WRDS SEC Analytics Suite
Control Variables		
Book to Market Value	Ratio of book value of equity to market value (Jiang et al., 2022).	Compustat
Firm Size (Log (Total Assets))	Natural log of total asset (Jiang et al., 2022).	Compustat
Leverage	Ratio of long-term debt to total assets (Jiang et al., 2022).	Compustat
ROA	Return on Assets (ROA) is defined as the ratio of Earnings Before Interests and Taxes (EBIT) to Total Asset (Harford & Li, 2007).	Compustat
R&D to Sales	Ratio of R&D to sales (Kamiya et al., 2021).	Compustat
Firm Age	Age of the firm (Jiang et al., 2022).	CRSP
Dividend Dummy	Dummy variable that takes the value of 1 if the firm pays dividend and 0 otherwise (Li, 2008).	Compustat
% Soft Assets	Percentage of assets that are neither cash nor property, plant, and equipment, calculated by (Total Assets − PP&E− Cash and Cash Equivalent)/Total Assets (Dechow et al., 2011).	Compustat
Abnormal Change in Employee Productivity	Abnormal change in employees as the percentage change in the number of employees less the percentage change in total assets (Dechow et al., 2011).	Compustat
Std. Deviation of ROA	Standard deviation of ROA over the last four years (following Harford & Li, 2007).	Compustat
Std. Deviation of Stock Return	Standard deviation of annual stock returns over the last four years (following Harford & Li, 2007).	CRSP

(*continued on next page*)

(*continued*)

Variable Name	Variable Description	Source
Big-4 Auditor	An indicator variable that takes the value of 1 if the auditor is a Big 4 auditor and 0 otherwise (Calderon and Gao, 2020).	Compustat
Percent of Independent Board Members	Proportion of board members who are independent (Cheng, 2008).	BoardEx
Board Size	Number of board members (Cheng, 2008).	BoardEx
Moderating Variables		
Cash to Total Assets	Income before extraordinary items plus depreciation and amortization divided by total assets (Kamiya et al., 2021). We use top and bottom terciles of these variables as moderators.	Compustat
Excess Stock Return	Buy-and-hold return for the year minus the CRSP value-weighted index return (Kamiya et al., 2021). We use top and bottom terciles of these variables as moderators.	CRSP

Appendix 2. Readability Indices used in this study

There are several approaches to measuring the readability of 10-K disclosures including the Fog Index, Bog Index, Flesch–Kincaid grade level and file size (Bonsall et al., 2017; Lo et al., 2017; Loughran & McDonald, 2014). Although the debates on the best measure of readability is still inconclusive, Asay, Libby, and Rennekamp (2018) opine that the Bog Index is less transparent because of its proprietary software while Bonsall et al. (2017) caution against the use of file size as a measure of readability because they find file size to be unstable and vary significantly over time largely because of formatting changes rather than the 10-K text itself. Consequently, we follow Lo et al. (2017) and mainly focus on the Fog Index in this study.

However, The Fog Index has been critiqued on the grounds that its inclusion of complex words could be problematical when used to assess readability of business documents that may, by nature, contain some complex words like corporation, company, telecommunications, etc. (Loughran & McDonald, 2014). In an attempt to mitigate this criticism of the Fog Index, Kim et al. (2019) modify the Fog Index to exclude commonly used multisyllable business words (in COMPUSTAT variable lists and Fama and French 49-industry descriptions) and find the Fog Index to be highly correlated with their Modified Fog Index (i.e., Pearson correlation of 0.901). We mitigate the criticism of the Fog Index by using the Kincaid Index as an alternative measure of readability in this study. Both the Fog Index and Kincaid Index are widely used in the literature (Hasan, 2020). Indeed, in noting the availability of tools to help fight complexity in disclosures "and the dense cover-your-assets boilerplate" disclosure documents, ex-SEC chairman Christopher Cox asserted that just "as the Black-Scholes model is a commonplace when it comes to compliance with the stock option compensation rules, we may soon be looking to the Gunning-Fog and Flesch-Kincaid models to judge the level of compliance with the plain English rules" (SEC, 2007). Below we present a brief discussion on Fog Index and Flesch-Kincaid Index.

Fog Index

The Fog Index, which indicates the number of years of formal education required to read text once and understand it (Li, 2008), is a combination of the number of words per sentence and the percent of complex words (Loughran & McDonald, 2014), scaled by the constant 0.4 in order to approximate years of formal education required to understand text on the first read (Lo et al., 2017). Specifically:

Fog Index = 0.4 * (words per sentence + percent of complex words).

Where words per sentence is defined as number of words divided by number of sentences; percent of complex words is defined as number of complex words divided by number of words multiplied by 100; and complex words are defined as words with more than two syllables.

The intuition behind the Fog Index is that documents with more syllables per word and/or more words per sentence are harder to read (Li, 2008). Lower values of the Fog Index indicate that a document is more readable (Loughran & McDonald, 2014). According to Li (2008), Fog Index greater than 18 means the document is unreadable, Fog Index of 14–18 means the document is difficult to read, Fog Index of 12–14 is ideal, Fog Index of 10–12 mean the document is acceptable and Fog Index of 8–10 means the document is childish.

Flesch-Kincaid formula or Kincaid Index

The Flesch-Kincaid formula, alternatively called the Kincaid Index, rates text by U.S. grade school level (Li, 2008), and is calculated as:

Kincaid Index = (11.8 * syllables per word) + (0.39 * words per sentence) - 15.59

Where syllables per word is defined as number of syllables divided by number of words and words per sentence is defined as number of words divided by number of sentences. A Kincaid Index of 9.0 signifies that an average ninth grader can understand the document (Laksmana, Tietz, & Yang, 2012).

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